Congruences for the Number of Subgroups of Order p^k in a Finite Group

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The method used by Serre [1, p. 147] to prove Sylow's theorem gives more general results, which we shall make explicit.

We use the following notation: p denotes a prime number; for every integer n, $v(n)$ is the exponent of the largest power of p dividing n; finally, G is a finite group of order g.

For completeness, recall the following lemma (Lazard [2, p. 71]).

Lemma 1. *a) If $n = \sum c_i p^i$ with $0 \le c_i < p$ (the expansion in base p),*

$$v(n!) = \frac{1}{p-1}\left(n - \sum c_i\right).$$

b)

$$v\binom{r}{s} \ge v(r) - v(s).$$

We prove a). One has

$$v(p^k!) = \sum_{i<k} i(p^{k-i} - p^{k-i-1}) = \sum_{i=0}^{k-1} p^i = \frac{p^k-1}{p-1},$$

which gives the assertion in this case; it is clear that if $0 < c < p$, then

$$v((cp^k)!) = c\,v(p^k!)$$

and that if $m < p^{v(n)}$, then

$$v((n+m)!) = v(n!) + v(m!).$$

These additivity properties are the same as those of the right-hand side; the formula follows.

Part b) follows easily by returning to the definitions and taking account of the algorithm for computing the digits c_i of a sum from those of its terms. There is equality if r is a power of p.

Proposition 1.

$$\binom{p^k n}{p^k m} \equiv \binom{n}{m} \pmod{p^{v(n)+2}}$$

if $p \neq 2$, or if n is odd. The congruence is always true modulo $p^{v(n)+1}$.

Proof. Let X and Y be two indeterminates:

$$(X+Y)^{p^k} = X^{p^k} + Y^{p^k} + pQ,$$

where Q is a polynomial of degree less than p^k in each variable. The binomial formula then shows that

$$(X+Y)^{p^k n} = (X^{p^k} + Y^{p^k})^n + np\,(X^{p^k} + Y^{p^k})^{n-1} Q \tag{1}$$

modulo the largest power of p which divides all the $p^i\binom{n}{i}$ $(i \ge 2)$. These have valuation at least

$$i + v(n) - v(i) \ge v(n) + 2$$

if $p \neq 2$; if $p = 2$, one always has $i > v(i)$ (Cantor), hence the exponent $v(n)+1$. The coefficient of $X^{p^k m} Y^{p^k(n-m)}$ in the left-hand side is $\binom{p^k n}{p^k m}$. The second term on the right-hand side of (1) contains no term of this type (no exponent occurring there is divisible by p^k), and the coefficient of this monomial in the first term on the right-hand side of (1) is therefore $\binom{n}{m}$. This proves the assertion if $p \neq 2$.

If $p = 2$, the preceding argument already proves

$$Q = X^{2^{k-1}} Y^{2^{k-1}} \pmod 2; \qquad Q^2 = X^{2^k} Y^{2^k} \pmod 4,$$

and the next term in the binomial expansion gives

$$\binom{2^k n}{2^k m} \equiv \binom{n}{m} + 2n(n-1)\binom{n-2}{m-1} \pmod{2^{v(n)+2}},$$

which proves the assertion.

Theorem. *Let S be a Sylow p-subgroup of G, let $s = v(g)$, and let d_k be the number of subgroups of G of order p^{s-k}, for $0 \leq k \leq s$. Then:*

(i) $d_k \equiv 1 \pmod p$*;*

(ii) if S is cyclic or abelian of type $(p, \ldots, p)$, then d_k is congruent modulo p^{k+1} to the number of subgroups of S of index p^k;

(iii) if S is not cyclic, then

$$d_1 \equiv 1 + p \pmod{p^2}.$$

Let the group G act by left translations on the set of its subsets with p^k elements. If such a subset P of G has stabilizer H, then it is a union of right cosets of H, and H is therefore a group of order p^ℓ with $\ell \leq k$; $\ell = k$ if and only if P is a translate (on the right or on the left, which is the same thing) of a subgroup of order p^k. The number of elements in the orbit of P is divisible by $p^{s-\ell}$. Thus $\binom{g}{p^k}$ is congruent modulo p^{s-k+1} to the number of subsets of G which are translates of a subgroup of order p^k, and by Proposition 1 it is also congruent to gp^{-k}. There are $d_{s-k} g p^{-k}$ such subsets, and

$$gp^{-k} \equiv d_{s-k} g p^{-k} \pmod{p^{s-k+1}},$$

which proves the first assertion.

We now consider case (ii), with S abelian of type $(p, \ldots, p)$. The number G_{ij} of subgroups of order p^i in a subgroup of order p^j is the number of points of a Grassmannian. For $i > j$, $G_{ij} = 0$; for $i \leq j$,

$$G_{ij} = \frac{(p^j - 1)\cdots(p^j - p^{i-1})}{(p^i - 1)\cdots(p^i - p^{i-1})} = G_i G_{j-i} G_j^{-1},$$

where

$$G_i = \left[(p^i - 1)\cdots(p-1)\right]^{-1} \quad \text{if } i \geq 0, \qquad G_i = 0 \quad \text{if } i < 0.$$

It is known that p^i divides the number u_i of subsets of G with p^s elements whose stabilizer has order p^{s-i}. Put $g = p^s h$. By counting in two different ways the number of pairs consisting of a subgroup of order p^{s-i} and a subset with p^s elements that it fixes, one obtains, for $0 \leq i, j \leq s$,

$$\begin{cases} d_i \dbinom{p^i h}{p^i} = \sum_j G_{s-i,s-j} u_j & (2) \\ p^i \mid u_i & (3) \\ d_s = 1. & (4) \end{cases}$$

We now work in the localization of $\mathbf{Z}$ at p (or in $\mathbf{Z}_p$). Let e_i be the quotient of the left-hand side of (2) by $\binom{p^i h}{p^i} G_i$. By Proposition 1 and by the fact that G_i is prime to p for $i \geq 0$, there exist numbers v_i, $0 \leq i \leq s$, such that

$$\begin{cases} d_i = G_i e_i \pmod{p^{i+1}} & (5) \\ e_i = \sum_j G_{i-j} v_j & (2') \\ p^i \mid v_i & (3') \\ e_s = G_s^{-1}. & (4') \end{cases}$$

Analogous identities are satisfied in the group S. To show that they determine $d_i \bmod p^{i+1}$, it is enough to check that the identities (for $0 \leq i, j \leq s$)

$$\begin{cases} e_i = \sum_j G_{i-j} v_j & (2'') \\ p^i \mid v_i & (3'') \\ e_s = 0 & (4'') \end{cases}$$

imply $p^{i+1} \mid e_i$. From (2'') one obtains

$$e_i - e_{i+1} = \sum_j (G_{i-j} - G_{i+1-j}) v_j.$$

Since trivially $G_k^{-1} = G_{k+1}^{-1} \pmod{p^{k+1}}$, one also has

$$p^{i+1-j} \mid G_{i-j} - G_{i+1-j} \quad \text{and} \quad p^{i+1} \mid e_i - e_{i+1}.$$

We conclude by (4'').

The same technique, in a simpler form, applies if S is cyclic.

It remains to consider case (iii). The subgroups of S of index p are all normal. Their intersection S' is the Fratini subgroup of S. For every subgroup T of S, if $TS' = S$, then $T = S$. In particular, if S is not cyclic, then S/S' is not cyclic either, and the subgroups of index p of S are identified with the hyperplanes of a $\mathbf{Z}/p$-vector space of dimension > 1; their number is congruent to $1 + p \pmod{p^2}$.

From the preceding equations there remains

$$\begin{cases} d_0 h = u_0, \\ d_1 \binom{ph}{p} = (1+p) u_0 + u_1 \pmod{p^2}, \\ \binom{ph}{p} = u_0 + u_1 \pmod{p^2}. \end{cases}$$

But by Proposition 1,

$$\binom{ph}{p} \equiv \binom{h}{1} = h \pmod{p^2},$$

and hence

$$d_1 h = (1+p) d_0 h + (h - d_0 h) \pmod{p^2}$$

and

$$d_1 = 1 + p d_0 \pmod{p^2}.$$

The assertion follows, since $d_0 \equiv 1 \pmod{p}$.

References

[1] J.-P. Serre, *Corps locaux*. Act. sc. et ind., Hermann, 1962.

[2] M. Lazard, *Groupes analytiques p-adiques*. Publ. Math. I.H.E.S., no 26, 1965.